



POLITECNICO
MILANO 1863

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Group 8

FINANCIAL ENGINEERING

Assignment n°3 BS

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1 Part I: Covariance Estimation

1.1 a. Calibration of Risk Aversion Parameter

The calibration of the risk aversion parameter γ on the first complete window available, which is from 31/12/2014 (reasonable, since we ask for 2 years of data from 1/1/2013) and 46 out of 50 assets passed the minimum coverage filter. In order to obtain a gross exposure close to 3 for the mean-variance ptf, the value of γ is equal to **180.16**. This kind of value guarantees a gross exposure equal to 3. We found this value using an added function which was not present in the initial files of utilities folder, this function takes advantage of the fact that the gross-exposure decreases in γ , which guarantees the existence and uniqueness of the risk aversion parameter given clearly the fact that the target gross exposure of the mean-variance ptf has to be not less than the one of the minimum variance ptf.

1.2 b. & c. Shrinkage Implementation

In the function is implemented the shrinkage which is a method who mitigates estimation error by computing a weighted average between the noisy sample covariance matrix and a structured target. The Constant Correlation model builds this target by assuming all asset correlations are equal to their mean, effectively filtering out statistical noise. Alternatively, the Single Factor Model structures the target around a single market factor. Errors found in the code:

1. line 35: missing sqrt to compute standard deviation
2. line 74-240 : the difference was computed with + instead of -
3. line 95: during the computation of rho_hat there was *2 instead of *0.5
4. line 104-105: the normalization of pi and rho was in the loop
5. line 150: to compute the intensity there wasn't T at the den
6. line 156: wrong formula to compute the new cov matrix
7. line 188: market variance computed with .std instead of .var
8. line 243: there wasn't += in the cycle but just =
9. line 268: pi wasn't normalized

1.3 d. Portfolio Computation

We computed the weights for both the minimum variance and the mean-variance portfolios. To properly evaluate the impact of the different estimation techniques, this computation was performed on three different covariance matrices: the sample covariance matrix and the two shrunk matrices (Constant Correlation and Single Factor). To correctly manage the Euro Stoxx 50 constituents, which change over time, the weights (NaNs) for stocks unavailable or excluded from the index at specific monthly rebalancing dates were set to zero using the .fillna(0.0) method. The **minimum variance portfolio** is obtained by solving:

$$\min_w w^\top \hat{\Sigma} w \quad \text{subject to} \quad \mathbf{1}^\top w = 1 \quad (1)$$

yielding the closed-form solution $w^* \propto \hat{\Sigma}^{-1} \mathbf{1}$. The **mean-variance portfolio** instead maximizes the risk-adjusted return:

$$\max_w w^\top \mu - \frac{\lambda}{2} w^\top \hat{\Sigma} w \quad (2)$$

giving $w^* \propto \hat{\Sigma}^{-1} \mu$, where λ is the risk aversion parameter calibrated in point 1a to target a gross exposure of approximately 3.

A visual inspection of the resulting weight vectors already suggests meaningful differences across the three covariance estimators, which will be analyzed in detail in the following section.

1.4 e. Discussion: Shrinkage Estimators

Minimum Variance Portfolios:

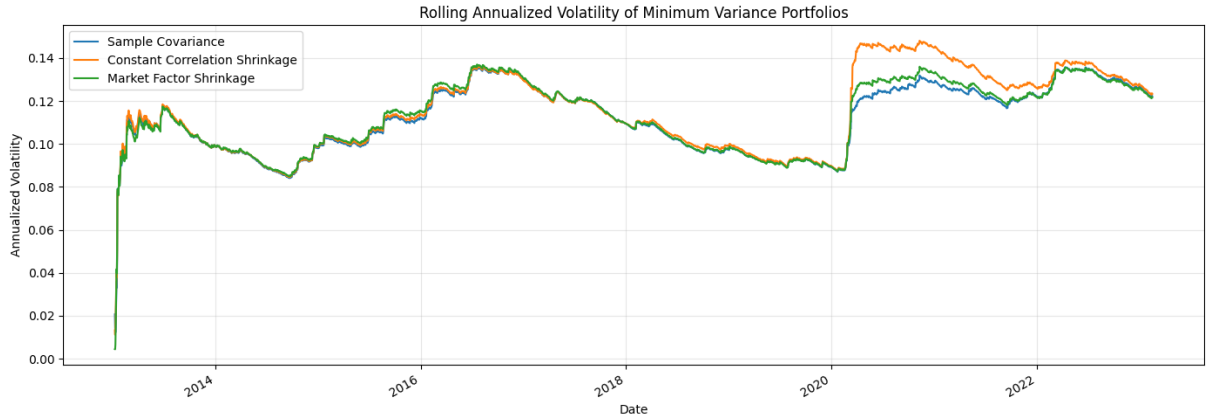


Figure 1: Volatility of Minimum Variance Portfolios

During the pre-COVID period, the three estimators produce almost identical results, with volatility moving steadily between 9% and 13%. The 2020 shock is the most revealing moment: all three portfolios experience a sharp jump, but the Constant Correlation Shrinkage reacts more violently than the others. This suggests it is more sensitive to the correlation shocks that usually happen during a crisis. After 2020, volatility does not return to its previous levels but stays structurally above 12%.

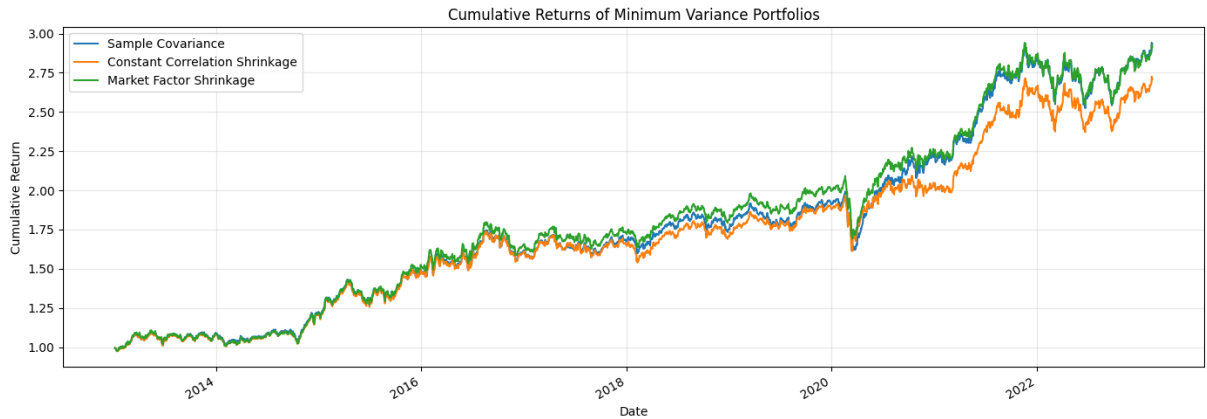


Figure 2: Cumulative Returns of Minimum Variance Portfolios

As before, the path up to 2020 is basically the same for everyone: the three estimators grow almost on top of each other, reaching returns of about 1.9x with nearly identical patterns. The COVID shock causes a sharp but short fall in value, followed by a quick recovery starting in the second half of 2020. After 2020, the Constant Correlation Shrinkage grows more slowly than the other two. It ends at about 2.6x, while the Market Factor Shrinkage and Sample Covariance both reach almost 3x and stay very close to each other.

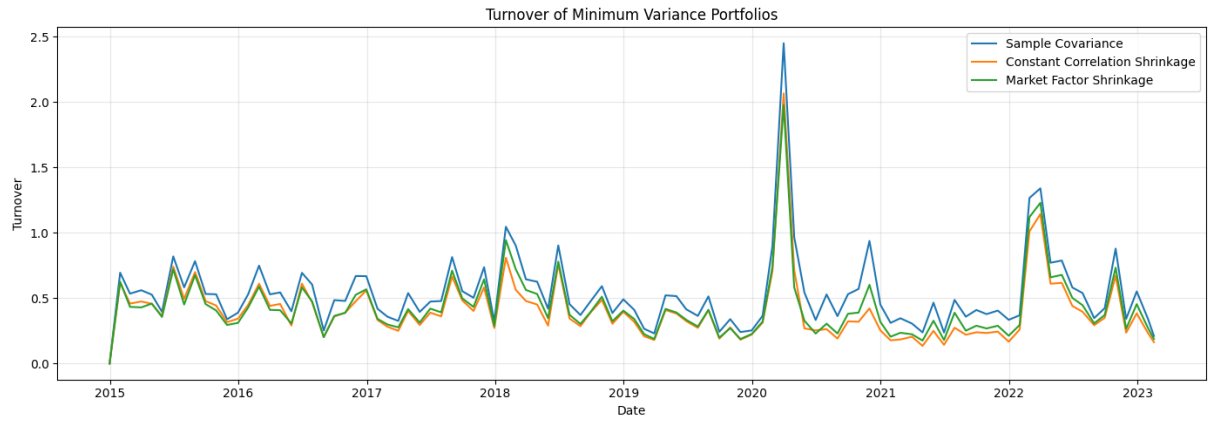


Figure 3: Turnover of Minimum Variance Portfolios

In normal conditions, the three estimators move in a similar way, staying between 0.2 and 0.8. However, the Sample Covariance is always higher than the other two. This is expected: the shrinkage methods create more stable results, which means they don't need many big or aggressive changes. During stressful times, like the 2020 peak and the market troubles in 2022, these differences grow. In these moments, the Sample Covariance reaches much higher levels compared to the shrinkage models.

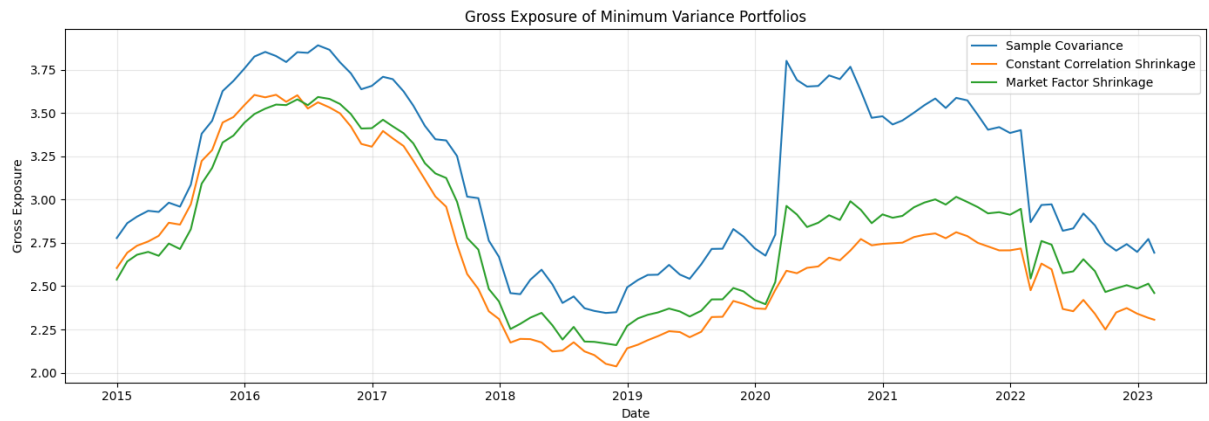


Figure 4: Gross Exposure of Minimum Variance Portfolios

The most obvious point is the clear gap between the Sample Covariance and the two shrinkage models. The Sample Covariance always leads to higher total exposure, peaking around 3.8-3.9, while the shrinkage models stay at lower levels. Between 2017 and 2019, all three move down together. However, they split apart again after 2020, with the Constant Correlation Shrinkage settling at the lowest levels of the entire period.

Mean-Variance Portfolios:

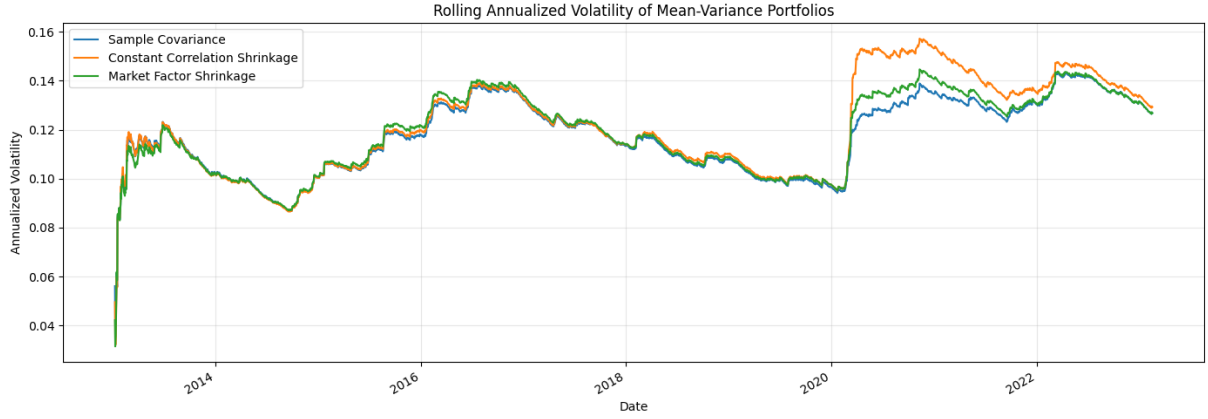


Figure 5: Volatility of Mean-Variance Portfolios

The overall pattern is very similar to the minimum variance portfolio: the three estimators stay almost the same during normal times, and differences only start to appear after 2020. Here too, the Constant Correlation Shrinkage is the most reactive during the COVID shock. It hits peaks of around 15.5%, which are slightly higher than those seen in the minimum variance case.

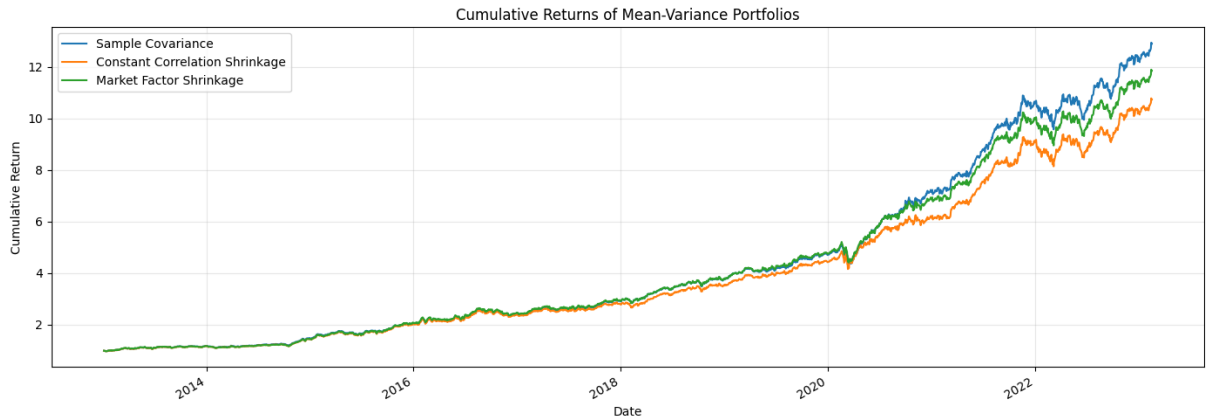


Figure 6: Cumulative Returns of Mean-Variance Portfolios

The graph shows the total returns of the mean-variance portfolios, and the difference compared to the minimum variance is clear: the performance is much higher, with the three models ending the period between 10x and 13x. As before, the Constant Correlation method gives the lowest returns. The highest returns come from the Sample Covariance approach, which even beats the Market Factor strategy. We believe this happens because shrinkage estimators smooth out the matrix toward a more uniform structure. While this provides more stability, it also means giving up some of the potential upside.

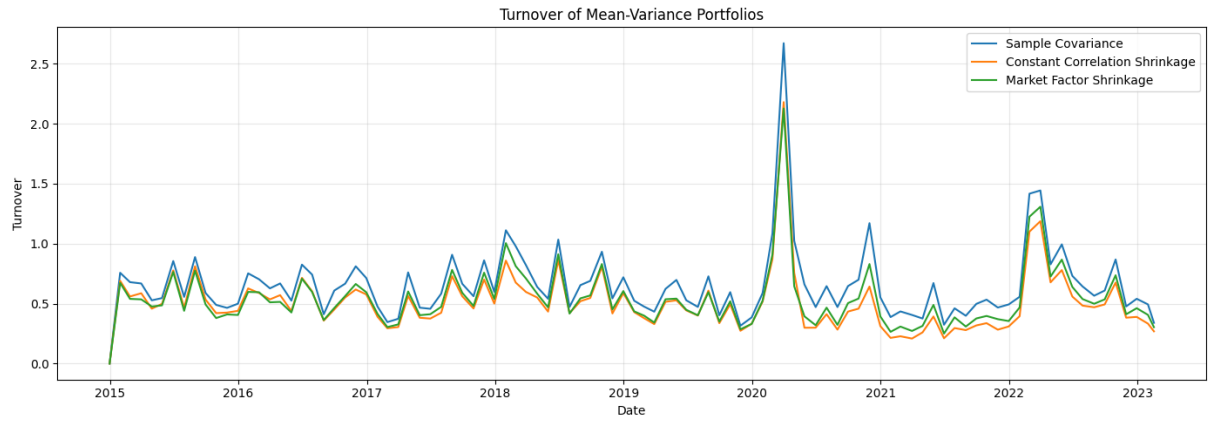


Figure 7: Turnover of Mean-Variance Portfolios

The mean-variance approach naturally creates a slightly higher turnover than the minimum variance one. However, the ranking of the estimators and how they react to shocks remain consistent across both approaches.

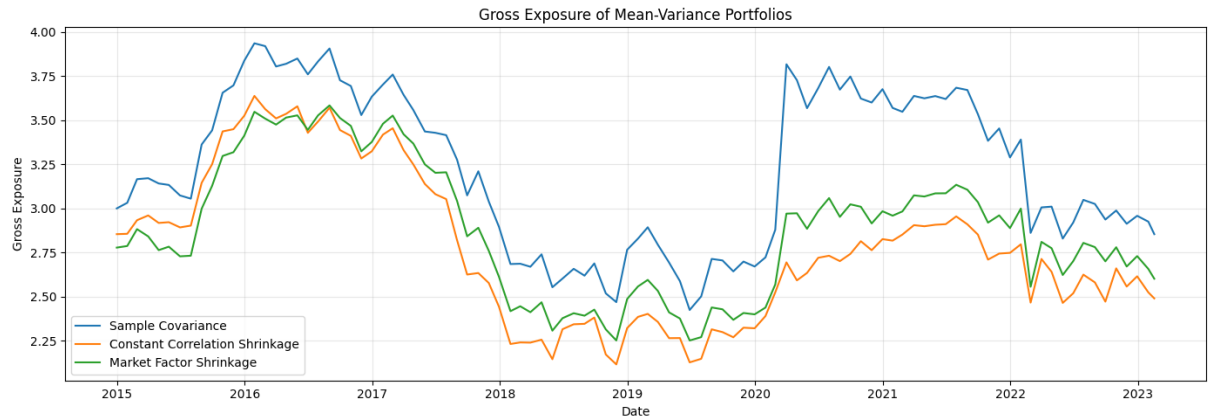


Figure 8: Gross Exposure of Mean-Variance Portfolios

The trend is similar to the Minimum Variance case, with a clear gap once again between the Sample Covariance and the two shrinkage models.

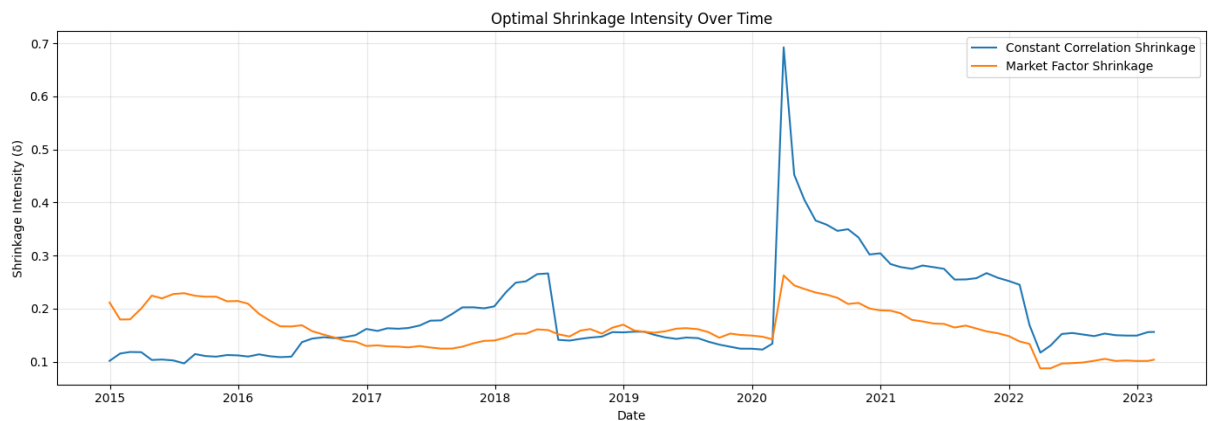


Figure 9: Optimal Shrinkage Intensity Over Time

In normal times, the two estimators behave differently: the intensity connected with the Market Factor Shrinkage starts higher and tends to decrease over time, while the one connected with the Constant Correlation Shrinkage starts lower and grows slowly until 2018-2019, when the two meet and almost overlap. The most revealing moment is the 2020 peak. The Constant Correlation Shrinkage jumps drastically to almost 0.70. This shows that, during the crisis, the algorithm felt it was necessary to rely much more on the target and much less on the raw sample data. The Market Factor Shrinkage curve reacts much more calmly, peaking around 0.25. This suggests that the market factor model remains a more stable way to represent covariance, even during stressful periods. After 2020, both return to lower values, though the Constant Correlation stays structurally higher than before.

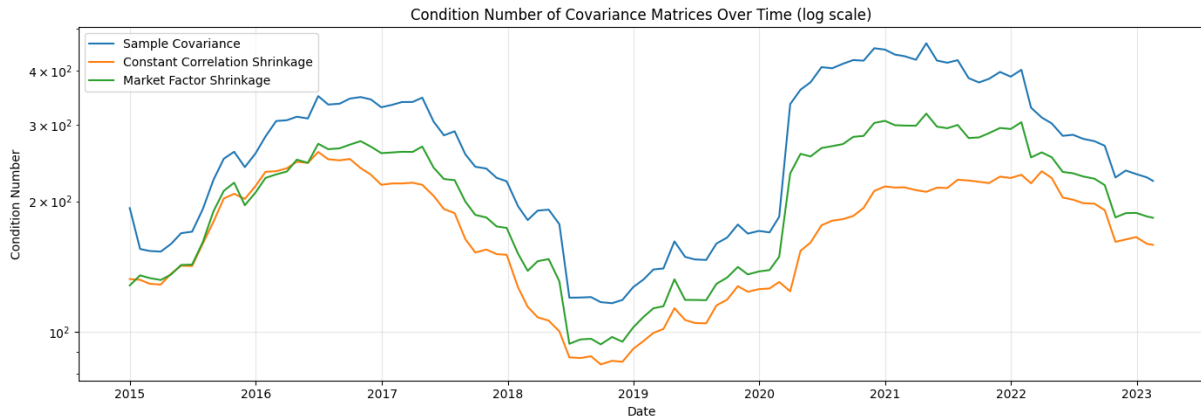


Figure 10: Condition Number of Covariance Matrices Over Time

The Sample Covariance is always the most unstable, with values constantly higher than the other two models and a sharp peak around 400 between 2020 and 2022. This explains much of what we saw in the previous graphs: higher turnover, more exposure, and a stronger reaction to shocks. The two shrinkage methods significantly reduce this instability. The Constant Correlation Shrinkage is the most stable of all, hitting lows of around 70-80 in 2018-2019. The Market Factor Shrinkage sits in the middle, behaving more like the Sample Covariance during stressful times. All three see a peak after 2020, but the intensity is very different. This confirms that shrinkage is very effective at keeping the matrix stable exactly when markets are turbulent and the data is "noisy."

2 Part II: Statistical Factor Model

2.1 a. & b. Principal Component Analysis (PCA) and Results Discussion

Factor Turnover, quantified via the *cosine similarity* between principal components at successive time intervals, serves as a proxy for the structural persistence of market drivers:

- **PC1: Market Factor** The *cosine similarity* of the first principal component (PC1) remains consistently near unity. This empirical evidence confirms that the Market Factor is the only structurally invariant component of the system. Regardless of the economic regime, PC1 acts as a common force coordinating the cohesive movement of securities.

- **Secondary PCs:** In contrast, secondary components (PC2-PC4) exhibit sharp and frequent drops in similarity, often falling below the critical threshold of 0.9. These *drops* signal a reconfiguration of the risk hierarchy: a driver that explains the variance of specific sectors at $t - 1$ (e.g., the technology sector represented by ASML or SAP) may be abruptly superseded at t by new macroeconomic or geopolitical factors, forcing a rotation of component weights and altering the nature of idiosyncratic risks.

The dynamics of the proportion of principal component needed to explain the variance provide critical insights into the shifting nature of market risk. During periods of strong trends or systemic crises, the proportion of PCs required to explain total variance contracts towards its historical minimum of approximately 12.24%. This phenomenon was most pronounced during the early 2020 COVID-19 shock, when the sudden collapse of global markets forced the system into an effectively one-dimensional state; in such phases, cross-asset correlations fall drastically and the entire market reacts synchronously to a single, dominant exogenous shock. Conversely, during periods devoid of global turmoil, such as January 2019, this proportion expands toward 30%, signaling a fragmentation of risk where securities follow idiosyncratic or sector-specific dynamics. This state of maximum dispersion reflects a market environment where firm-specific fundamentals prevail over a unified global macroeconomic direction.

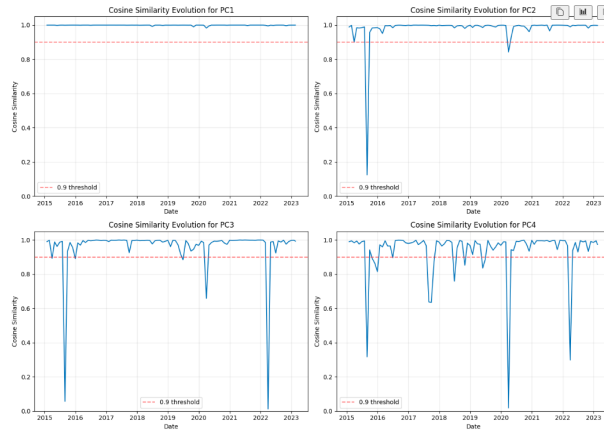


Figure 11: Cosine Similarity Evolution

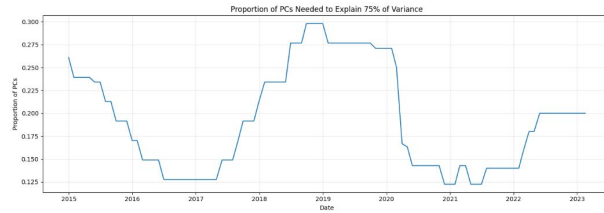


Figure 12: Proportion of PCs Needed to Explain 75% of Variance

The selection of the estimation window poses a critical trade-off between responsiveness and stability. An excessively short time window makes the model highly reactive to recent shifts but introduces significant instability; this leads to an inflation of factor turnover driven by statistical noise, giving life to unjustified and frequent rotations of the principal components. On the other hand, an excessively long window enhances factor

stability. In this scenario, the model fails to provide timely signals regarding market changes, as the preponderance of historical data reduces the impact of new information.

Correlation vs. Covariance Matrix: Utilizing the correlation matrix standardizes the data, thereby neutralizing the influence of varying volatility scales across different securities. Through the covariance matrix, the Principal Components would be dominated by assets or sectors with intrinsically higher volatility. Correlation, by contrast, facilitates an analysis based strictly on the structure of linear dependencies between assets. Given the heterogeneity of the underlying components, ranging from high-growth technology stocks like *Adyen/ASML* to stable utilities such as *Enel/Iberdrola*, the use of the correlation matrix should be preferred to ensure a balanced and scale-invariant factor decomposition.

c. Interpretation of the First Three Factors as of January 31, 2019

1. **PC1 - Market Factor:** This represents the systematic risk of the Eurozone. This factor shows significant positive loadings across almost all 50 components, heavily influenced by index heavyweights such as *SAP* (Information Technology), *LVMH* (Consumer Discretionary), and *TotalEnergies* (Energy). It is the primary driver capturing the overall direction of the European equity market.
2. **PC2 - Financials vs. Growth:** The second factor reflects the divergence between the Financials sector (represented by *Santander*, *BNP Paribas*, *Intesa Sanpaolo*, and *Allianz*) and Growth sectors. In early 2019, this component captured market sensitivity to the "low-for-longer" interest rate environment and the flattening of the yield curve, which pressured bank margins while supporting the high valuations of tech-oriented stocks like *ASML*.
3. **PC3 - Industrials & Materials vs. Defensive:** The third component showed significant loadings in defensive sectors such as Consumer Staples and Health Care, reflecting the European industry's exposure to global trade tensions and manufacturing demand. This factor, which captures the intention of investors to shelter in low-volatility, non-cyclical assets to hedge against geopolitical uncertainty, isolates heavy cyclical components (Industrials such as Siemens, Schneider Electric, Airbus and Materials such as Air Liquide) from defensive sectors like Health Care (Sanofi, Bayer) and Utilities (Enel, Iberdrola).

3 Part III: Connecting the Dots

3.1 a. Shrinkage vs PCA-based Denoising

On January 31st, 2019—a period marked by significant market dispersion—both methodologies converge toward the objective of regularizing the covariance matrix. Shrinkage stabilizes the estimator by preventing the smallest eigenvalues from collapsing toward zero, effectively blending the sample matrix with a structured target. Simultaneously, PCA denoising isolates the systematic risk by preserving the k largest eigenvalues (the signal) and replacing the remaining $N-k$ components with their average. This combined approach ensures that the risk representation of the Euro Stoxx 50 remains robust, avoiding the overfitting of idiosyncratic noise while maintaining the integrity of the global

matrix structure. The empirical behavior of the spectrum at the end of January 2019 demonstrates that as the number of retained principal components increases from 1 to 10, the PCA model gradually captures more granular market dynamics at the expense of potential noise inclusion. With $k = 1$, the model is overly parsimonious, capturing only the systemic Market Factor; as k moves toward 10, the model begins to incorporate specific sectoral links (Financials, Technology, Industrials). Shrinkage complements this by ensuring that the "tail" of the spectrum—the eigenvalues not captured by the top 10 PCs—is not left unregularized. By comparing the two, it is evident that shrinkage provides a continuous regularization across the entire spectrum, whereas PCA denoising provides a discrete approximation. For the Euro Stoxx 50 on this specific date, the high proportion of PCs required to explain 75% of the variance suggests that a very low k (e.g., 1 or 3) might omit significant structural information, while the shrinkage-stabilized matrix ensures that even when more components are considered, the resulting covariance remains well-conditioned and less prone to the erratic factor turnover typical of the raw sample matrix.

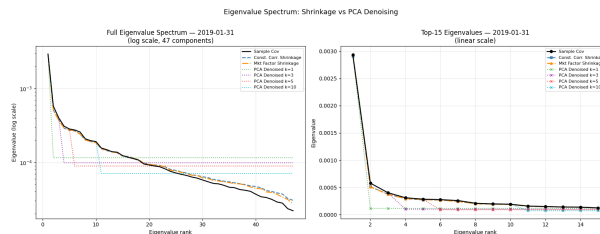


Figure 13: Eigenvalue Spectrum: Shrinkage vs PCA Denoising

3.2 b. Limited Cross-Sectional Dimension

While linear shrinkage is a sophisticated theoretical tool, its practical impact within the Euro Stoxx 50 dataset remains constrained due to the limited cross-sectional dimension ($N = 50$). Since the number of assets is small relative to the available time-series observations (T), the N/T ratio is sufficiently low. This ensures that the sample covariance matrix S is already sufficiently stable and does not suffer from the severe numerical degradation or ill-conditioning that shrinkage is designed to correct in much larger universes (such as the S&P 500 or STOXX 600).

This structural limitation is further compounded by the homogeneous volatility profiles of the index's constituents. As the Euro Stoxx 50 tracks highly liquid, large-cap blue-chip stocks, the statistical "noise" in the tail of the eigenvalue spectrum is minimal. Consequently, regularizing the matrix toward the Identity target yields only marginal adjustments, as there are no extreme distortions or massive estimation errors to rectify.